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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING (ARTIFICIAL INTELLIGENCE)

Academic Year: 2026-2027

Batch No: A3

PROJECT TITLE: "An Intelligent Cloud Resource and Cost Optimization Framework using Machine Learning and Explainable AI"

ABSTRACT:

Modern cloud-native environments frequently experience significant financial inefficiencies due to over-provisioned resources, idle infrastructure, unattached storage volumes, and the absence of proactive cloud cost management. In addition to operational waste, organizations are increasingly exposed to economic security threats such as Denial-of-Wallet (DoW) attacks, where malicious workloads intentionally trigger excessive cloud resource consumption, resulting in unnecessary financial expenditure. Existing FinOps solutions primarily provide historical billing visualization and rule-based monitoring but offer limited capabilities for predictive analytics, intelligent decision support, explainable recommendations, and secure remediation.

To address these challenges, this framework proposes an intelligent cloud resource and cost optimization framework that integrates Machine Learning and Explainable Artificial Intelligence (XAI) to support proactive cloud operations. The proposed framework employs a multi-model analytical pipeline comprising Prophet for cloud cost forecasting, XGBoost for business-aware unit economics prediction, and Isolation Forest for cloud anomaly and Denial-of-Wallet detection. The generated predictions are further interpreted using Explainable AI techniques to provide transparent recommendations, confidence scores, estimated cost savings, risk assessment, and root-cause analysis, enabling engineering teams to make informed optimization decisions.

To reduce manual operational effort, the framework incorporates an Infrastructure-as-Code (IaC) recommendation engine capable of automatically generating production-ready Terraform templates and cloud command-line configurations for resource optimization. Machine Learning (ML), Explainable Artificial Intelligence (XAI), Cloud Computing, Cloud Resource Optimization, Cloud Cost Optimization, FinOps, Infrastructure-as-Code (IaC), Denial-of-Wallet (DoW) Detection.To ensure operational safety and governance, all infrastructure modifications follow a Human-in-the-Loop (HITL) workflow, where generated remediation actions require administrator review and approval before execution. Furthermore, the framework supports real-time notifications through enterprise collaboration platforms and provides an AI-powered conversational assistant that enables users to interact with cloud infrastructure using natural language, understand optimization recommendations, and receive contextual explanations of prediction results.

The proposed framework is expected to improve cloud resource utilization, accurately forecast cloud expenditure, detect cloud cost anomalies and Denial-of-Wallet attacks, reduce manual cloud management effort, and enhance operational transparency, financial efficiency, and decision-making in enterprise cloud environments.

Keywords: Machine Learning (ML), Explainable Artificial Intelligence (XAI), Cloud Computing, Cloud Resource Optimization, Cloud Cost Optimization, FinOps, Infrastructure-as-Code (IaC), Denial-of-Wallet (DoW) Detection.

SYSTEM SPECIFICATIONS:

H/W Specifications:

Processor	Intel Core i5 Processor (or equivalent)
RAM	16 GB
Hard Disk	256 GB SSD (Minimum)

S/W Specifications

Operating System	Windows 10/11 or Linux (Ubuntu) [Cross-platform deployment supported via Docker]
Server-side Script	FastAPI (Python-based asynchronous backend framework)
Languages and Technologies	Python, TypeScript, SQL, React.js, Tailwind CSS, Machine Learning, Explainable AI (XAI), Large Language Models (LLMs), LangGraph, AI Agents, Model Context Protocol (MCP), Cloud Computing, FinOps, Cloud Security (SecOps), AWS Cloud Services
IDE	Visual Studio Code (VS Code) , Google Colab
Libraries Used	FastAPI, scikit-learn, XGBoost, Prophet, SHAP, Pandas, NumPy, SQLAlchemy, Pydantic, LangChain, LangGraph, Plotly, boto3 (AWS SDK), Requests, Uvicorn, Psycopg2

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